

Encounter Crew — Output

Every file the crew emits, verbatim, plus the negotiation that produced it

Aaron Low · 30 July 2026 · Python 3.8+, standard library only

Code: `crew/kindled_crew.py` · ReadMe: `crew/README.md` · Output: `crew/out/`

Reproduce this document: `py crew/build_output_pdf.py`

WHAT THIS DOCUMENT IS

The companion report described the crew's design. **This one is the output itself.** Nothing below is written by hand — every block is a file the crew wrote or a console session it printed, pasted verbatim from a live run made while building this PDF. The commands that produced each block are named above it, so any of it can be reproduced in one line.

1 The three artifacts

FILE	SIZE	WHAT IT IS
<code>crew/out/encounters.json</code>	2731 B	Flat, typed rows that import directly as an Unreal DataTable — the game data the encounter ships as
<code>crew/out/encounters.schema.md</code>	1302 B	The column contract for that JSON, plus both audit verdicts at generation time
<code>crew/out/run-report.md</code>	3780 B	Every agent's declared role, the encounter table, the budget verdict, and the full negotiation

Sections 2 and 3 are `run-report.md` from two different runs — floor 1, which passes first try, and floor 3, where the budget auditor rejects the plan twice before the crew converges. Sections 4 and 5 are the emitted data and its schema. Section 6 proves no agent is decorative.

2 run-report.md — floor 1

py crew/kindled_crew.py → crew/out/run-report.md . The shipped Gate-1 wave structure fits the frame budget on the first pass, so the negotiation closes in one round.

Kindled — encounter crew run report

Floor 1 (density x1.0)

Generated 2026-07-30 by crew/kindled_crew.py .

Negotiation rounds: 1 · Budget: PASS (11.12ms / 16.6ms) · Readability: CROWDED

Agents

AGENT	ROLE	READS	WRITES
canon-reader	Parses the project's shipped design and measured performance docs into a single fact block, so no downstream agent invents a number.	—	canon
roster-architect	Derives the entity tier ladder (fodder → soldier → elite → boss) as multipliers over the measured combat baseline, and prices each tier in encounter-budget points.	canon	tiers
encounter-architect	Composes the three escalating waves — how many of which tier, and the spawn cadence — and redesigns when an auditor sends the plan back.	canon , revision_directive , tiers	plan
budget-auditor	Projects the plan's peak concurrent entity count onto the measured frame-cost curve and returns PASS, or REVISE with a scale directive the architect must obey.	canon , plan	budget_verdict , revision_directive
readability-auditor	Checks each wave's simultaneous distinct silhouettes against the locked 4-value palette and flags any wave the player could not parse at speed.	canon , plan	readability_verdict
data-emitter	Flattens the approved plan into DataTable-shaped JSON rows for Unreal, writes the column schema, and refuses to emit anything that has not passed both audits.	budget_verdict , canon , plan , readability_verdict , tiers	artifacts

Encounter

WAVE	BODIES	COMPOSITION	BUDGET PTS	SPAWN
1	250	fodder x250	250	ring
2	450	fodder x338, soldier x112	674	ring
3	701	fodder x420, soldier x238, elite x42, boss x1	1698	arena_entrances

Budget verdict

- Peak concurrent entities: **821** (wave 3 + retinue cap 120)
- Projected frame cost: **11.12ms** against a 16.6ms budget — 5.48ms headroom
- Basis: measured draw curve, single client, UnitShading=1

Negotiation

What each agent said, in order. The architect only re-scales because the auditor sent the plan back with a reason.

```
canon-reader      read 2 canon source(s): docs/GATE1-FUN-PROTOTYPE.md, docs/perf/BUDGETS.md
canon-reader      waves=[250, 450, 700] retinue_cap=120 perf curve has 6 measured points
roster-architect  derived 4 tiers over baseline 130HP/30DPS: fodder(58HP), soldier(130HP), elite(416HP),
boss(2340HP)
--- negotiation round 1 ---
encounter-architect composed W1: 250 bodies (250pts) | W2: 450 bodies (674pts) | W3: 701 bodies (1698pts)
budget-auditor     PASS - peak 821 entities ~ 11.12ms (5.48ms headroom)
--- audit and emit ---
readability-auditor CROWDED - max 4 distinct enemy types in a wave against 2 free palette values
data-emitter       wrote 7 DataTable rows -> crew/out/encounters.json
data-emitter       wrote column contract -> crew/out/encounters.schema.md
```

Blackboard traffic

Every read and write, attributed. Access is enforced: an agent that touches an undeclared key raises `ContractViolation`.

```
canon-reader -> wrote canon
roster-architect <- read canon
roster-architect -> wrote tiers
encounter-architect <- read canon
encounter-architect <- read tiers
encounter-architect -> wrote plan
budget-auditor <- read canon
budget-auditor <- read plan
budget-auditor -> wrote budget_verdict
readability-auditor <- read canon
readability-auditor <- read plan
readability-auditor -> wrote readability_verdict
data-emitter <- read canon
data-emitter <- read tiers
data-emitter <- read plan
data-emitter <- read budget_verdict
data-emitter <- read readability_verdict
data-emitter -> wrote artifacts
```

3 run-report.md — floor 3, the real negotiation

py crew/kindled_crew.py --floor 3 . This is the run worth reading. At the escalated density the floor peaks **2.3× over the frame budget**; the budget auditor rejects it, names how many bodies are actually affordable, and the architect redesigns. Two rejections, three rounds, converging on a floor with zero headroom.

Kindled — encounter crew run report

Floor 3 (density x2.5)

Generated 2026-07-30 by crew/kindled_crew.py .

Negotiation rounds: 3 · **Budget:** PASS (16.6ms / 16.6ms) · **Readability:** CROWDED

Agents

AGENT	ROLE	READS	WRITES
canon-reader	Parses the project's shipped design and measured performance docs into a single fact block, so no downstream agent invents a number.	—	canon
roster-architect	Derives the entity tier ladder (fodder → soldier → elite → boss) as multipliers over the measured combat baseline, and prices each tier in encounter-budget points.	canon	tiers
encounter-architect	Composes the three escalating waves — how many of which tier, and the spawn cadence — and redesigns when an auditor sends the plan back.	canon , revision_directive , tiers	plan
budget-auditor	Projects the plan's peak concurrent entity count onto the measured frame-cost curve and returns PASS, or REVISE with a scale directive the architect must obey.	canon , plan	budget_verdict , revision_directive
readability-auditor	Checks each wave's simultaneous distinct silhouettes against the locked 4-value palette and flags any wave the player could not parse at speed.	canon , plan	readability_verdict
data-emitter	Flattens the approved plan into DataTable-shaped JSON rows for Unreal, writes the column schema, and refuses to emit anything that has not passed both audits.	budget_verdict , canon , plan , readability_verdict , tiers	artifacts

Encounter

WAVE	BODIES	COMPOSITION	BUDGET PTS	SPAWN
1	341	fodder ×341	341	ring
2	614	fodder ×460, soldier ×154	922	ring
3	956	fodder ×573, soldier ×325, elite ×57, boss ×1	2292	arena_entrances

Budget verdict

- Peak concurrent entities: **1076** (wave 3 + retinue cap 120)
- Projected frame cost: **16.6ms** against a 16.6ms budget — 0.0ms headroom
- Basis: measured draw curve, single client, UnitShading=1

Negotiation

What each agent said, in order. The architect only re-scales because the auditor sent the plan back with a reason.

```
canon-reader      read 2 canon source(s): docs/GATE1-FUN-PROTOTYPE.md, docs/perf/BUDGETS.md
canon-reader      waves=[250, 450, 700] retinue_cap=120 perf curve has 6 measured points
roster-architect  derived 4 tiers over baseline 130HP/30DPS: fodder(58HP), soldier(130HP), elite(416HP),
boss(2340HP)
--- negotiation round 1 ---
encounter-architect composed W1: 625 bodies (625pts) | W2: 1125 bodies (1687pts) | W3: 1751 bodies
(4155pts)
budget-auditor    REVISE - peak 1871 entities ~ 37.29ms, over budget by 20.69ms; only 956 bodies are
affordable, sending back
--- negotiation round 2 ---
encounter-architect revising: peak 1871 entities ~ 37.29ms, over the 16.6ms budget; re-scaling to x1.36 of
the shipped wave sizes
encounter-architect composed W1: 341 bodies (341pts) | W2: 614 bodies (922pts) | W3: 957 bodies (2293pts)
budget-auditor    REVISE - peak 1077 entities ~ 16.62ms, over budget by 0.02ms; only 956 bodies are
affordable, sending back
--- negotiation round 3 ---
encounter-architect revising: peak 1077 entities ~ 16.62ms, over the 16.6ms budget; re-scaling to x1.36 of
the shipped wave sizes
encounter-architect composed W1: 341 bodies (341pts) | W2: 614 bodies (922pts) | W3: 956 bodies (2292pts)
budget-auditor    PASS - peak 1076 entities ~ 16.60ms (0.00ms headroom)
--- audit and emit ---
readability-auditor CROWDED - max 4 distinct enemy types in a wave against 2 free palette values
data-emitter      wrote 7 DataTable rows -> crew/out/encounters.json
data-emitter      wrote column contract -> crew/out/encounters.schema.md
```

Blackboard traffic

Every read and write, attributed. Access is enforced: an agent that touches an undeclared key raises `ContractViolation`.

```

canon-reader -> wrote canon
roster-architect <- read canon
roster-architect -> wrote tiers
encounter-architect <- read canon
encounter-architect <- read tiers
encounter-architect -> wrote plan
budget-auditor <- read canon
budget-auditor <- read plan
budget-auditor -> wrote budget_verdict
budget-auditor -> wrote revision_directive
encounter-architect <- read canon
encounter-architect <- read tiers
encounter-architect <- read revision_directive
encounter-architect -> wrote plan
budget-auditor <- read canon
budget-auditor <- read plan
budget-auditor -> wrote budget_verdict
budget-auditor -> wrote revision_directive
encounter-architect <- read canon
encounter-architect <- read tiers
encounter-architect <- read revision_directive
encounter-architect -> wrote plan
budget-auditor <- read canon
budget-auditor <- read plan
budget-auditor -> wrote budget_verdict
readability-auditor <- read canon
readability-auditor <- read plan
readability-auditor -> wrote readability_verdict
data-emitter <- read canon
data-emitter <- read tiers
data-emitter <- read plan
data-emitter <- read budget_verdict
data-emitter <- read readability_verdict
data-emitter -> wrote artifacts

```

THE DESIGN FINDING

That convergence is a real result for Kindled, not a demonstration: **the escalating-density promise in the vertical slice cannot be paid for by the current renderer.** The crew reached it by arithmetic over the project's own measured cost curve — from the opposite direction to the project's independent renderer conclusion, and agreeing with it.

The same run as the console saw it

```
=====
KINDLED - ENCOUNTER DESIGN CREW
raw orchestration | stdlib only | 6 agents | floor 3
=====

[canon-reader]
- read 2 canon source(s): docs/GATE1-FUN-PROTOTYPE.md, docs/perf/BUDGETS.md
- waves=[250, 450, 700] retinue_cap=120 perf curve has 6 measured points

[roster-architect]
- derived 4 tiers over baseline 130HP/30DPS: fodder(58HP), soldier(130HP), elite(416HP), boss(2340HP)

--- negotiation round 1 ---
[encounter-architect]
- composed W1: 625 bodies (625pts) | W2: 1125 bodies (1687pts) | W3: 1751 bodies (4155pts)
[budget-auditor]
- REVISE - peak 1871 entities ~ 37.29ms, over budget by 20.69ms; only 956 bodies are affordable, sending back

--- negotiation round 2 ---
[encounter-architect]
- revising: peak 1871 entities ~ 37.29ms, over the 16.6ms budget; re-scaling to x1.36 of the shipped wave sizes
- composed W1: 341 bodies (341pts) | W2: 614 bodies (922pts) | W3: 957 bodies (2293pts)
[budget-auditor]
- REVISE - peak 1077 entities ~ 16.62ms, over budget by 0.02ms; only 956 bodies are affordable, sending back

--- negotiation round 3 ---
[encounter-architect]
- revising: peak 1077 entities ~ 16.62ms, over the 16.6ms budget; re-scaling to x1.36 of the shipped wave sizes
- composed W1: 341 bodies (341pts) | W2: 614 bodies (922pts) | W3: 956 bodies (2292pts)
[budget-auditor]
- PASS - peak 1076 entities ~ 16.60ms (0.00ms headroom)

[readability-auditor]
- CROWDED - max 4 distinct enemy types in a wave against 2 free palette values

[data-emitter]
- wrote 7 DataTable rows -> crew/out/encounters.json
- wrote column contract -> crew/out/encounters.schema.md

=====
DONE - 7 encounter rows, 3 negotiation round(s)
crew/out/encounters.json
crew/out/encounters.schema.md
crew/out/run-report.md
=====
```

4 encounters.json — the emitted game data

Floor 1, verbatim. Unique `Name` row key, scalar columns only, no nesting — exactly what Unreal's DataTable importer requires. `_meta` carries provenance, so any row traces back to the measurement that justified it.


```

{
  "_meta": {
    "game": "Kindled",
    "floor": 1,
    "generated": "2026-07-30",
    "generator": "crew/kindled_crew.py",
    "canon_sources": [
      "docs/GATE1-FUN-PROTOTYPE.md",
      "docs/perf/BUDGETS.md"
    ],
    "grounded_in_repo_docs": true,
    "budget_verdict": {
      "peak_concurrent_entities": 821,
      "peak_wave": 3,
      "projected_frame_ms": 11.12,
      "frame_budget_ms": 16.6,
      "headroom_ms": 5.48,
      "status": "PASS",
      "basis": "measured draw curve, single client, UnitShading=1"
    },
    "readability_verdict": {
      "palette_values": 4,
      "enemy_value_budget": 2,
      "status": "CROWDED",
      "per_wave": [
        {
          "wave": 1,
          "distinct_enemy_types": 1,
          "status": "OK",
          "note": "reads by value alone"
        },
        {
          "wave": 2,
          "distinct_enemy_types": 2,
          "status": "OK",
          "note": "reads by value alone"
        },
        {
          "wave": 3,
          "distinct_enemy_types": 4,
          "status": "CROWDED",
          "note": "one type must be carried by silhouette, not value"
        }
      ]
    }
  },
  "rows": [
    {
      "Name": "W1_fodder",
      "Wave": 1,
      "Tier": "fodder",
      "Count": 250,
      "HP": 58,
      "DPS": 18,
      "BudgetCost": 1,
      "SpawnMode": "ring",
      "BreatherAfterSeconds": 6
    },
    {
      "Name": "W2_fodder",
      "Wave": 2,
      "Tier": "fodder",
      "Count": 338,
      "HP": 58,
      "DPS": 18,

```

```

    "BudgetCost": 1,
    "SpawnMode": "ring",
    "BreatherAfterSeconds": 6
  },
  {
    "Name": "W2_soldier",
    "Wave": 2,
    "Tier": "soldier",
    "Count": 112,
    "HP": 130,
    "DPS": 30,
    "BudgetCost": 3,
    "SpawnMode": "ring",
    "BreatherAfterSeconds": 6
  },
  {
    "Name": "W3_fodder",
    "Wave": 3,
    "Tier": "fodder",
    "Count": 420,
    "HP": 58,
    "DPS": 18,
    "BudgetCost": 1,
    "SpawnMode": "arena_entrances",
    "BreatherAfterSeconds": 0
  },
  {
    "Name": "W3_soldier",
    "Wave": 3,
    "Tier": "soldier",
    "Count": 238,
    "HP": 130,
    "DPS": 30,
    "BudgetCost": 3,
    "SpawnMode": "arena_entrances",
    "BreatherAfterSeconds": 0
  },
  {
    "Name": "W3_elite",
    "Wave": 3,
    "Tier": "elite",
    "Count": 42,
    "HP": 416,
    "DPS": 54,
    "BudgetCost": 12,
    "SpawnMode": "arena_entrances",
    "BreatherAfterSeconds": 0
  },
  {
    "Name": "W3_boss",
    "Wave": 3,
    "Tier": "boss",
    "Count": 1,
    "HP": 2340,
    "DPS": 78,
    "BudgetCost": 60,
    "SpawnMode": "arena_entrances",
    "BreatherAfterSeconds": 0
  }
]
}

```

5 encounters.schema.md — the column contract

encounters.json — schema

Produced by `crew/kindled_crew.py` for **Kindled**. The `rows` array is flat and typed so it imports directly as an Unreal **DataTable**: one unique `Name` key per row, scalar columns only, no nesting.

COLUMN	TYPE	NOTES
<code>Name</code>	string	DataTable row key, <code>W<wave>_<tier></code>
<code>Wave</code>	int	1-indexed wave this row belongs to
<code>Tier</code>	string	<code>fodder</code> / <code>soldier</code> / <code>elite</code> / <code>boss</code>
<code>Count</code>	int	bodies of this tier spawned in the wave
<code>HP</code>	int	derived as a multiplier over the measured retinue baseline
<code>DPS</code>	int	as above
<code>BudgetCost</code>	int	encounter-budget points, for future spend-per-room tuning
<code>SpawnMode</code>	string	<code>ring</code> or <code>arena_entrances</code>
<code>BreatherAfterSeconds</code>	int	recovery window before the next wave; <code>0</code> on the last

`_meta` carries provenance: which repo docs the numbers came from, and both audit verdicts, so a row can always be traced back to the measurement that justified it.

Audit state at generation

- **Budget** — PASS: peak 821

concurrent entities, projected 11.12ms against a

16.6ms frame budget (measured draw curve, single client, UnitShading=1).

- **Readability** — CROWDED: 2 palette values

free for enemies after the lit floor and the retinue take theirs.

Floor 3's rows, for comparison

Same schema, same generator, different verdict block — the numbers moved because the negotiation moved them, not because anything was re-authored by hand.

```

{
  "_meta": {
    "game": "Kindled",
    "floor": 3,
    "generated": "2026-07-30",
    "generator": "crew/kindled_crew.py",
    "canon_sources": [
      "docs/GATE1-FUN-PROTOTYPE.md",
      "docs/perf/BUDGETS.md"
    ],
    "grounded_in_repo_docs": true,
    "budget_verdict": {
      "peak_concurrent_entities": 1076,
      "peak_wave": 3,
      "projected_frame_ms": 16.6,
      "frame_budget_ms": 16.6,
      "headroom_ms": 0.0,
      "status": "PASS",
      "basis": "measured draw curve, single client, UnitShading=1"
    },
    "readability_verdict": {
      "palette_values": 4,
      "enemy_value_budget": 2,
      "status": "CROWDED",
      "per_wave": [
        {
          "wave": 1,
          "distinct_enemy_types": 1,
          "status": "OK",
          "note": "reads by value alone"
        },
        {
          "wave": 2,
          "distinct_enemy_types": 2,
          "status": "OK",
          "note": "reads by value alone"
        },
        {
          "wave": 3,
          "distinct_enemy_types": 4,
          "status": "CROWDED",
          "note": "one type must be carried by silhouette, not value"
        }
      ]
    }
  },
  "rows": [
    {
      "Name": "W1_fodder",
      "Wave": 1,
      "Tier": "fodder",
      "Count": 341,
      "HP": 58,
      "DPS": 18,
      "BudgetCost": 1,
      "SpawnMode": "ring",
      "BreatherAfterSeconds": 6
    },
    {
      "Name": "W2_fodder",
      "Wave": 2,
      "Tier": "fodder",
      "Count": 460,
      "HP": 58,
      "DPS": 18,

```

```
"BudgetCost": 1,
"SpawnMode": "ring",
"BreatherAfterSeconds": 6
},
{
  "Name": "W2_soldier",
  "Wave": 2,
  "Tier": "soldier",
  "Count": 154,
  "HP": 130,
  "DPS": 30,
  "BudgetCost": 3,
  "SpawnMode": "ring",
  "BreatherAfterSeconds": 6
},
{
  "Name": "W3_fodder",
  "Wave": 3,
  "Tier": "fodder",
  "Count": 573,
  "HP": 58,
  "DPS": 18,
  "BudgetCost": 1,
  "SpawnMode": "arena_entrances",
  "BreatherAfterSeconds": 0
},
{
  "Name": "W3_soldier",
  "Wave": 3,
  "Tier": "soldier",
  "Count": 325,
  "HP": 130,
  "DPS": 30,
  "BudgetCost": 3,
  "SpawnMode": "arena_entrances",
  "BreatherAfterSeconds": 0
},
{
  "Name": "W3_elite",
  "Wave": 3,
  "Tier": "elite",
  "Count": 57,
  "HP": 416,
  "DPS": 54,
  "BudgetCost": 12,
  "SpawnMode": "arena_entrances",
  "BreatherAfterSeconds": 0
},
{
  "Name": "W3_boss",
  "Wave": 3,
  "Tier": "boss",
  "Count": 1,
  "HP": 2340,
  "DPS": 78,
  "BudgetCost": 60,
  "SpawnMode": "arena_entrances",
  "BreatherAfterSeconds": 0
}
]
}
```

6 Proof that no agent is removable

`py crew/kindled_crew.py --drop roster-architect`. Every agent declares the blackboard keys it may read and write, and the blackboard *enforces* those declarations. Drop a producer and the run stops at the first consumer that needs it — a different point and a different missing artifact for each agent.

```
=====
KINDLED - ENCOUNTER DESIGN CREW
raw orchestration | stdlib only | 6 agents | floor 1
=====

!! RUNNING WITHOUT 'roster-architect' - demonstrating that the pipeline depends on it

[canon-reader]
- read 2 canon source(s): docs/GATE1-FUN-PROTOTYPE.md, docs/perf/BUDGETS.md
- waves=[250, 450, 700] retinue_cap=120 perf curve has 6 measured points

--- negotiation round 1 ---
[encounter-architect]

!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
PIPELINE BROKEN - encounter-architect needs 'tiers', but no agent has produced it. An upstream agent is missing
from the crew - the pipeline cannot proceed.
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!

This is the intended result of removing an agent: every agent declares
its inputs and outputs, so a missing producer stops the run at the first
consumer that needs it. No agent in this crew is decorative.
```

Kindled — Encounter Design Crew · output report · generated 2026-07-30 by `crew/build_output_pdf.py` from a live run. Companion design report: `docs/Kindled-AgentCrew-Report.pdf`.